

# MOHAMMED AMAN BHUIYAN

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## EXPERIENCE

### Simplex Infocom LLC

Sep '25 – Present

AI/ML (Engineer)

Delaware, USA

- Built end-to-end **ASR** and **TTS** pipelines for a global **IELTS Speaking assessment platform**, enabling real-time speech evaluation and automated feedback generation.
- Prototyped and evaluated state-of-the-art ASR models and speech-to-speech models (e.g., **Moshi**, **Hibiki**) to improve conversational latency and naturalness in AI-driven interactions.
- Reduced inference latency and improved throughput by deploying optimized pipelines using **ONNX**, **TensorRT**, and **CTranslate2 (C++)** for large-scale speech models.
- Presented system architecture, model training processes, and AI roadmap in client-facing meetings, technical decisions into product impact.
- Worked in an **Agile** development environment, contributing to sprint planning, task execution, and continuous delivery of features.

### North South University (NSU)

Feb '24 – May '26

Graduate Teaching Assistant (Department of ECE)

Dhaka, Bangladesh

- Instructed **CSE445: Machine Learning** (Graduate level), guiding hands-on model implementation, and supporting research-oriented coursework.
- Co-taught **CSE373: Design & Analysis of Algorithms** (Undergraduate), mentoring 100+ students in algorithmic problem-solving, complexity analysis, and programming techniques.
- Designed assignments, evaluated coursework, and held consultation hours to improve students conceptual clarity and academic performance.

### Research Project – Classroom Monitoring Technology

Jan '25 – Sep '25

(Worked Under the supervision of Dr. Shafin Rahman)

North South University

- Developed an **end-to-end AI pipeline** aligned with thesis research for automated classroom analytics using real-world video data.
- Trained and optimized deep learning models on a proprietary classroom dataset provided by the supervisor.
- Built a full-stack application that processes classroom video input and generates comprehensive engagement reports.
- Integrated data preprocessing, model inference, evaluation metrics, and reporting into a production-ready system.

## EDUCATION

### North South University

Sep '21 – Dec '25

Bachelor of Science in Computer Science and Engineering, CGPA: **3.68 (Magna Cum Laude)**

Dhaka, Bangladesh

## TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C++
- **AI & ML Tools:** PyTorch, HuggingFace, Scikit-learn, LangChain, LlamaIndex, Ollama
- **Data Manipulation & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Web:** FastAPI, Flask, MERN Stack, Jest, JSDoc
- **Databases:** MySQL, MongoDB, PostgreSQL, Redis, VectorDB (FAISS, ChromaDB, Qdrant)
- **Other Expertise:** Git, Docker, REST API Development, Data Analysis, Web Scraping, RAG Pipelines, LLM/SLM Fine-tuning, Model Compression, Prompt Engineering

## PROJECTS

### Educational Chatbot using RAG

🔄 Physics\_RAG\_Chatbot

SLMs, Python, Ollama, LangChain, HuggingFace, FastAPI, Prompt Engineering, Git

- Developed a lightweight Retrieval-Augmented Generation (RAG) chatbot using small language models (SLMs) for offline and efficient educational assistance
- Built pipeline with **LangChain**, **Ollama**, and **HuggingFace** for document retrieval, embedding, and response generation
- Evaluated performance on a **5,000+** self-curated physics Q&A dataset, achieving **75%** response accuracy using **RAGChecker** framework
- Designed a user-friendly web interface with **FastAPI** to improve accessibility for students in low-resource environments

## SHOBDOTORI: Regional-Dialect-to-Standard Bangla Speech Recognition

 ShobdoTori\_ASR

*Python, PyTorch, Whisper, Wav2Vec2, KenLM, ASR*

- Developed an Automatic Speech Recognition (ASR) system to transcribe regional Bangladeshi dialects into standard Bangla.
- Curated a **3,800+** audio-sample dataset covering **20 regional** dialects across Bangladesh.
- Implemented Transformer-based ASR (**Whisper, Wav2Vec2**) with phoneme alignment and audio data augmentation.
- Applied **n-gram KenLM** post-processing to improve transcription fluency and linguistic correctness.
- Evaluated performance using Normalized Levenshtein Similarity (**NLS**), achieving competitive leaderboard results.

## BanglaWhisperDiar: Long-Form ASR & Speaker Diarization for Bangla

 BitwiseMind\_ASR\_Diar

*Python, PyTorch, Whisper, PyAnnote, PyTorch Lightning, librosa*

- Built a dual-pipeline deep learning system for long-form Bengali conversational audio transcription and speaker diarization.
- Curated a **15,000+** aligned audio-segment dataset using sequential fuzzy text matching with active transcription, and applied stochastic audio augmentations to improve model robustness.
- Fine-tuned the **Whisper-Medium** ASR model using 8-bit AdamW optimization and mixed-precision training for efficient large-scale training.
- Optimized long-audio inference with **FasterWhisper CT2Translate**, processing a 24+ hour test set in **25 minutes** (Only **1 minute** per audio file).
- Fine-tuned the **PyAnnote Segmentation 3.0** model using PyTorch Lightning and integrated it into a customized speaker diarization pipeline.
- Evaluated cross-domain generalization using Word Error Rate (WER) and Diarization Error Rate (DER) as benchmarking metrics, achieving a **28.4%** improvement in WER and a **40.3%** improvement in DER compared to the baseline.

## TimeClipAI: Real-Time Action Classification and Time Segmentation

 TimeClip\_AI

*PyTorch, Anchor Transformers, I3D, Python, Git*

- Developed **TimeClipAI**, a framework for **online temporal action localization** using Anchor Transformers
- Enabled **real-time action classification and time segmentation** in full-length videos
- Integrated support for multiple benchmark datasets including EGTEA, EPIC-Kitchen 100, THUMOS'14, and CricShot10
- Leveraged pre-trained **I3D features** for efficient training and testing
- Designed and trained a **post-processing Offset Scoring Network (OSN)** to refine predictions
- Achieved **state-of-the-art** performance with robust generalization across datasets

## Publications

### Z-Pruner: Post-Training Pruning of Large Language Models for Efficiency without Retraining

Samiul Basir Bhuiyan, Md. Sazzad Hossain Adib, Mohammed Aman Bhuiyan, Muhammad Rafsan Kabir, Moshir Farazi, Shafin Rahman, and Nabeel Mohammed

Accepted at **IEEE AICCSA 2025**  [arxiv.org/abs/2508.15828](https://arxiv.org/abs/2508.15828)

### DUGO: Dual-Scale Gaussian Memory for Online Temporal Action Localization in Egocentric Videos

Mohammed Aman Bhuiyan, Md. Sazzad Hossain Adib, Samiul Basir Bhuiyan, Mir Faiyaz Hossain, Syed Mohammed Shamsul Islam, Md Zakir Hossain, and Shafin Rahman

Under Review at **IEEE Big Data 2026**

## Extra-Curricular Activities & Competitions

### National AI Competition Finalist, AI Fication 2025 (Televerse 1.0)

Chittagong University of Engineering & Technology (CUET)

- Focused on regional-to-standard Bangla speech transcription
- Ranked **4th** in the private online round and **7th overall** in the final stage

### Finalist (out of 192+ teams), BUET AI HACKATHON DL Sprint 4.0

Bangladesh University of Engineering & Technology (BUET)

- Built solutions for Bangla long-form audio ASR transcription and speaker diarization
- Ranked **2nd** in the private online round and **8th overall** in the final stage

### Top 15 Finalist (out of 100+ teams), NSU Capstone Project Challenge – Innovation Challenge 2025

North South University (NSU)

- Recognized for developing impact-driven and innovative project solutions

## References

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